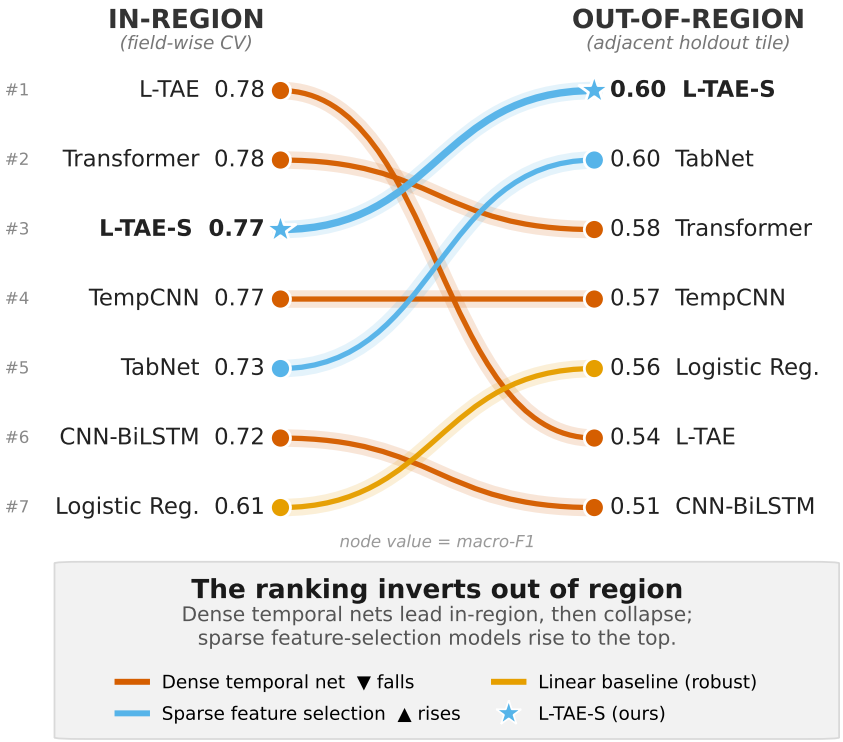
Graphical Abstract

Sparse Feature Selection Outweighs Deep Learning Capacity for Out-of-Region Transfer in Crop Classification

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Abstract

Crop-type maps from satellite image time series are a backbone of agricultural monitoring, yet the machine-learning models that produce them are almost always trained and tested within a single region, leaving open how they behave when deployed to new areas where no labels were collected. Using a South African Sentinel-2 dataset (optical only), we benchmark twenty-three classical machine-learning and deep-learning models under out-of-region transfer, applying them without any target-region labels or adaptation to an adjacent, non-overlapping holdout tile, and identify the model property most associated with success: sparse, axis-aligned feature selection. Models that commit to a few features at a time, gradient-boosted trees, logistic regression, and the attentive TabNet, transfer robustly (macro-F1 0.56–0.60), whereas high-capacity dense temporal networks (CNN-BiLSTM, a Transformer encoder, L-TAE) degrade sharply; in a controlled comparison, adding sparse feature selection to L-TAE improves transfer more than adding attention capacity does. We turn this insight into a method: L-TAE-S, a sparse-attention temporal encoder that ties the best model for out-of-region macro-F1 (0.60) at a fraction of the training cost, with a field-level variant that matches this accuracy while training in seconds per seed. We further show that conventional in-region validation inverts these rankings, the dense networks that lead in-region (macro-F1 0.72–0.78) fall furthest under transfer, which is why models intended for operational crop mapping should be evaluated outside the region they were trained on. Averaging pixels at the field level typically narrows the transfer gap, but also raised holdout accuracy for L-TAE, and the most transferable models retain near-full accuracy using only a quarter of the training fields. Together these results give practitioners a concrete design principle for crop classifiers that must be applied to nearby unlabeled farmland.

Keywords: Crop classification, deep learning, feature selection, remote sensing, Sentinel-2, out-of-region transfer, time-series classification

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1. Introduction

Accurate and timely crop classification is foundational to precision agriculture and to the agencies that monitor food security and agricultural land use, enabling yield forecasting, resource optimization, and food-security assessment (Bégué et al., 2020; Ienco et al., 2017). The Sentinel-2 constellation, with its multispectral richness and roughly five-day revisit, has made dense multi-temporal optical imagery widely available. Yet reliable crop classification remains hard throughout Africa and the Global South: spectral similarity between crops, irregular planting and phenological cycles, persistent cloud cover, and rain-fed dryland management all degrade accuracy (Bégué et al., 2020; Mahlayeye et al., 2024). Classical methods such as Random Forest (RF) and XGBoost have proven effective and efficient on engineered features (Mahlayeye et al., 2024; Mann et al., 2026; Mann and Warner, 2017; Mann et al., 2019), but their performance hinges on hand-crafted feature engineering (Maxwell et al., 2018; Belgiu and Drăguţ, 2016). Deep learning, recurrent and convolutional temporal models, reduces reliance on hand-crafted predictors and captures phenological structure that summary statistics miss (Zhu et al., 2017; Rußwurm and Körner, 2018; Pelletier et al., 2019), but its generalizability in complex, low-label settings remains uncertain (Pankajakshan et al., 2025).

A second, and for our purposes critical, gap concerns *how generalization is measured*. Much of the crop-classification literature reports accuracy under random or field-wise cross-validation drawn from a single region (Belgiu and Drăguţ, 2016; Garnot and Landrieu, 2020; Pelletier et al., 2019; Rußwurm and Körner, 2018; Ienco et al., 2017). Such protocols hold out individual pixels or fields but not *space*: training and test observations share the same atmosphere, soils, management calendars, and acquisition geometry, so the reported numbers reflect in-distribution fit rather than transfer to new terrain. Because operational maps are produced by applying a model to regions where no labels were collected, this distinction matters, yet studies that re-rank models under a spatially disjoint holdout remain rare (Pankajakshan et al., 2025; Jin et al., 2019; Rustowicz et al., 2019). This differs from the bulk of the transfer-learning literature, which adapts a model using target-domain data: labels, unlabeled imagery, or self-supervised pre-training (Ma et al., 2024; Wang et al., 2023; Mai et al., 2025). Our setting is the no-target-access case (domain generalization): models are trained on source tiles and applied to the holdout directly, with no adaptation, so what determines transfer is a property intrinsic to the model rather than a target-specific tuning step.

The Western Cape of South Africa concentrates the operational challenges, smaller fields, rain-fed dryland systems, heterogeneous planting rotations, and persistent winter cloud cover, while offering an unusually rich, high-quality labeled dataset (Radiant Earth Foundation, 2024). This makes it a realistic test bed: rich enough to train and compare modern architectures fairly, yet hard enough to expose the failure modes that matter in practice. We evaluate deep temporal models (CNN-BiLSTM, TabNet, L-TAE, a Transformer encoder, and TempCNN) and deep patch-based 2D/3D CNNs against classical XGBoost,

LightGBM, and RF. Critically, we evaluate every model twice: under field-wise cross-validation *within* the training region, and on an adjacent, non-overlapping holdout tile that no training pixel touches. We restrict inputs to multi-temporal Sentinel-2 optical imagery alone, to isolate the spectral-time-series signal and maximize applicability where only optical data are available. All code is public (Mann, 2025a).

This study makes five contributions. First, a multi-architecture benchmark (twenty-three classical, deep temporal, and deep patch-based models) showing that in-region versus out-of-region transfer evaluation produce *inverted* rankings on the same data. Second, empirical evidence that models with sparse, axis-aligned feature-selection, the property behind two decades of RF and gradient-boosted-tree dominance, are more stable when moved to a holdout tile, and TabNet inherits it via sparsemax masks. Third, a controlled architectural comparison, tested with paired bootstraps, in which adding sparse feature selection improves transfer more than adding attention capacity. We introduce L-TAE-S, a sparse-attention encoder that ties TabNet for the best out-of-region macro-F1, with a field-level variant that matches this accuracy while training in seconds per seed. Fourth, evidence that field-level aggregation is not a reliable substitute for feature selection. Aggregation narrows the transfer gap of dense models but raised holdout accuracy only for L-TAE. Fifth, a data-efficiency analysis under out-of-region transfer showing the most transferable models are also the most data-efficient, holding near-full holdout accuracy on a quarter of the training fields; as a supporting experiment, a controlled comparison of `xr_fresh` time-series features against raw reflectance for XGBoost and TabNet under out-of-region transfer. This leads to our central question: across classical machine-learning and deep-learning models, which most reliably classifies crops from multi-temporal Sentinel-2 optical imagery when applied, without target-region labels, to an adjacent region outside the training tiles, and what model properties determine whether in-distribution accuracy survives the transfer? Our aim is not to build a single operational model but to test how model families behave under the low-information conditions common across Africa: no target-region labels, a modest single-season training set, and optical imagery only.

1.1. Related Work

Early crop classification relied on classical learners (RF, SVM, logistic regression, gradient-boosted trees) operating on engineered descriptors (spectral indices, temporal summaries) valued for efficiency, interpretability, and robustness on structured data (Maxwell et al., 2018; Belgiu and Drăguţ, 2016; Breiman, 2001; Chen and Guestrin, 2016; Ke et al., 2017), though their performance depends on feature engineering that can limit transferability across agroecological regions (Wardlow et al., 2007; Mahlayeye et al., 2024). Deep models learn representations directly from reflectance sequences: CNNs capture local spectral-spatial pattern (Kussul et al., 2017), and recurrent and attention architectures model seasonal phenology (Ienco et al., 2017; Rußwurm and Körner, 2018; Garnot and Landrieu, 2020; Rußwurm and Körner, 2020). Other designs

combine convolutional feature extraction with temporal modeling (Fan et al., 2023; Tufail et al., 2025; Bandar and Coşkunçay, 2024), or use deep backbones as feature extractors feeding classical classifiers (Espejo-Garcia et al., 2020; Koklu et al., 2022; Yang et al., 2020). A recurring assumption is that richer learned representations generalize better than compact engineered ones; whether this holds under out-of-region transfer, rather than within-region validation, is less established and is a central question here. Architectures that explicitly target cross-scenario generalization (e.g., Cropformer (Wang et al., 2023)) remain the exception and are rarely benchmarked against classical and deep baselines on a common spatially disjoint holdout. We fill this gap with a systematic comparison under *both* protocols, and additionally evaluate `xr_fresh` as an automated time-series feature-extraction framework in a crop-classification setting (Mann, 2025b).

2. Data

The study uses a labeled field dataset distributed through Radiant Earth's MLHub (Radiant Earth Foundation, 2024) and used as the basis of the AI4FoodSecurity (South Africa) challenge (AI4EO, 2021), in an agriculturally productive part of the Western Cape (Fig. 1). Two adjacent Sentinel-2 tiles (`34S_19E_258N`, `34S_19E_259N`) form the training region (4,150 labeled fields over 622 km^2; mean field 15.0 ha); a third, immediately adjacent tile (`34S_20E_259N`) is reserved as a holdout for out-of-region transfer evaluation (2,417 fields over 282 km^2). The tiles do not overlap, so no pixel or field is shared, and transfer must contend with different soils, geometries, and acquisition conditions. Because the holdout lies within the same agroecological zone and growing calendar, this is a relatively *local* shift, and the gaps we report are best read as a conservative lower bound on degradation under broader transfer. Throughout, "in-region" denotes a field-wise train/validation/test split within the training region; "out-of-region" (OOR) or "holdout" denotes predictions on the adjacent, non-overlapping tile, made without any target-region labels or adaptation. We use "out-of-region" rather than the broader "spatial transfer" because the holdout tests a local shift, not transfer across agroecological zones or years.

We classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley, and small-grain grazing. The classes are heavily imbalanced; lucerne/medics is 43% of training fields and 56% of the holdout (Supplementary Material, Fig. S1(a)), which motivates macro-F1 as the primary metric (Sec. 3.3) and class-imbalance handling during training. Spectrally, wheat and barley overlap strongly, the dominant source of persistent confusion (Supplementary Material, Fig. S2(b)). Their seasonal profiles show the same pattern: lucerne/medics and canola have distinct phenology, whereas the three cereals follow nearly the same curve, and the holdout season runs later than in the training region (Supplementary Material, Sec. S-H, Fig. S8). Sentinel-2 Level-2A surface reflectance was acquired for the January–December 2017 growing season, cloud-masked with `s2cloudless` plus a shadow mask, and composited monthly. Six features per month are used: bands B2, B6, B11, B12, plus EVI and Hue, with EVI and

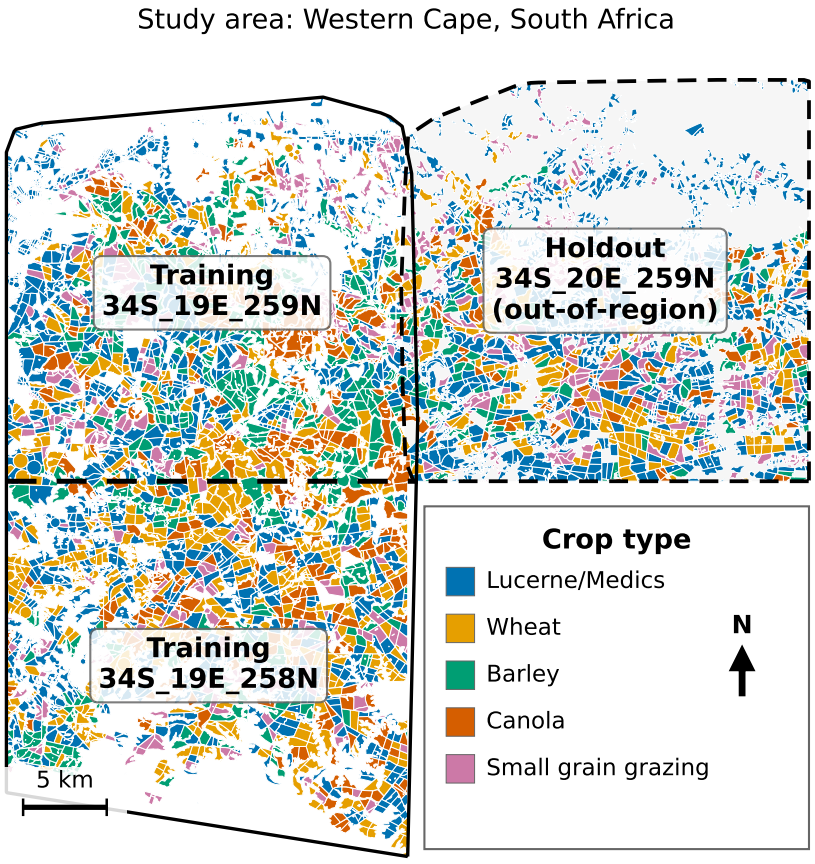


Figure 1: Study area in the Western Cape, South Africa. Two adjacent Sentinel-2 tiles (`34S_19E_258N`, `34S_19E_259N`) form the training region; the immediately adjacent `34S_20E_259N` tile (dashed) is held out for out-of-region transfer evaluation. Fields are colored by crop type; the tiles are spatially disjoint, so no pixel or field is shared between training and holdout.

Hue computed per scene before compositing, chosen to span the most informative spectral regions while limiting redundancy among correlated bands (Mann et al., 2026; Tufail et al., 2025). June 2017 yielded no usable imagery and May was discarded for cloud, leaving 10 monthly observations, so each pixel is a 60-dimensional time series (6 features $\times$ 10 months). Field polygons were digitized from sub-meter aerial photography and carry a single seasonal crop label (Radiant Earth Foundation, 2024); all predictions are made at the field level.

3. Methodology

We compare two model families: classical machine learning on engineered `xr_fresh` features (9 models) and deep learning (14), the latter comprising deep temporal models on per-pixel or per-field time series (12) and deep patch-based models on image patches (2). To prevent leakage, all train/validation/test splitting is performed at the field level with a fixed random seed, so no field's pixels straddle partitions; pixel- and patch-level predictions are aggregated to a field label by majority vote (or, for probabilistic models, mean pooling). Computational cost across paradigms is reported in the Supplementary Material (Sec. S-C).

3.1. Classical Machine Learning

We automate feature extraction with the `xr_fresh` toolkit, computing a set of statistical and temporal descriptors (e.g., autocorrelation, linear trend, day-of-year of the seasonal maximum, quantiles, complexity) from each pixel's 10-month profile, yielding a 174-dimensional per-pixel vector (Mann et al., 2026; Mann, 2025b); the full feature list is in the Supplementary Material (Sec. S-A, Table S1). Four baselines: logistic regression, RF (300 trees), LightGBM, and XGBoost (James et al., 2013; Breiman, 2001; Ke et al., 2017; Chen and Guestrin, 2016; Friedman, 2001) are trained on this representation, all of them tree- or margin-based with built-in feature selection or shrinkage; standard formulations are reproduced in the Supplementary Material (Sec. S-A). We also report soft-voting and stacked ensembles (RF/XGBoost/LightGBM, with logistic-regression meta-learner), an Optuna-tuned field-level XGBoost, and a SMOTETomek-resampled variant; the *Stacking* and *SMOTE Stacked* pair differs only in resampling, isolating the effect of class rebalancing. At the field level, pixel features are mean-pooled within each polygon before classification (Peña-Barragán et al., 2011).

3.2. Deep Learning Architectures

At the pixel level each observation is a 60-feature temporal sequence. Although the pixel models train on millions of pixels, pixels within a field are highly autocorrelated and share one label, so the number of statistically independent units is closer to the number of fields; field-wise splitting keeps evaluation honest at the field scale. We evaluate, as five-seed ensembles, a CNN-BiLSTM trained with weighted focal loss; TabNet (Arik and Pfister, 2021; Muruganantham et al., 2024), which treats the observations as tabular input and, at each decision step, uses an attentive transformer to produce a *sparse* feature mask via sparsemax, selecting a handful of features and ignoring the rest; the L-TAE (Garnot and Landrieu, 2020), which applies a single learned query over the temporal sequence; a standard multi-head Transformer encoder (Vaswani et al., 2017; Rußwurm and Körner, 2020); and TempCNN (Pelletier et al., 2019). Detailed equations for these standard models, and the field-level voting rule, are in the Supplementary Material (Sec. S-A). The Transformer is included as an informative dense-attention comparison: it increases attention capacity relative

to L-TAE while retaining no explicit sparse feature-selection mechanism. Deep patch-based models (a multi-channel 2D CNN and a 3D CNN) tile each field into 100 m patches resampled to 128×128 and are likewise trained as five-seed ensembles; details in the Supplementary Material (Sec. S-A).

3.2.1. Sparse-Attention L-TAE (L-TAE-S)

To test whether a tree-like, sparse inductive bias can be grafted onto a temporal-attention encoder, we introduce L-TAE-S, which inserts a TabNet-inspired channel-selection gate before the attention heads (Fig. 2). For each of the $n_{\text{head}} = 16$ heads, a per-head linear projection of the embedded, positionally-encoded sequence ($d_{\text{model}} = 128$, key dimension $d_k = 8$) is passed through *sparsemax* (Martins and Astudillo, 2016) to produce a sparse, instance-dependent mask over embedding channels. Formally, for head h and time step $\mathbf{z}_t \in \mathbb{R}^E$,

$$\mathbf{m}_{h,t} = \text{sparsemax}\big(\mathbf{P}_{h,t} \odot \text{BN}(W_h \mathbf{z}_t)\big), \quad \tilde{\mathbf{z}}_{h,t} = \mathbf{m}_{h,t} \odot \mathbf{z}_t,$$

$$\mathbf{P}_{h+1,t} = \mathbf{P}_{h,t} \odot \max(\gamma - \mathbf{m}_{h,t},\, 0), \qquad \mathbf{P}_{1,t} = \mathbf{1};$$

head h then forms its keys and values from $\tilde{\mathbf{z}}_{h,t}$ before applying L-TAE attention. Because *sparsemax* projects onto the probability simplex, most channels of $\mathbf{m}_{h,t}$ are exactly zero, and the relaxation prior $\mathbf{P}_{h,t}$ (coefficient $\gamma = 1.5$) discourages successive heads from re-selecting the same channels. These two mechanisms are the sole source of sparsity. The implementation also adds a penalty $\lambda \sum_{h,t} \|\mathbf{m}_{h,t}\|_1$ ($\lambda = 10^{-3}$), but because each sparsemax mask sums to one, this term is constant and has zero gradient, so it does not contribute to sparsity. This single block, the only architectural difference from the L-TAE, grafts a tree-like sparse feature-selection bias onto the attention backbone while keeping its temporal modeling, mirroring TabNet's sparsemax masks (Arik and Pfister, 2021). We evaluate L-TAE-S at both the pixel and field level as a five-seed ensemble under the same weighted-focal objective and field-aggregation protocol as the other temporal networks. Its sensitivity to γ, the number of heads, and an explicit sparsity penalty is reported in the Supplementary Material (Sec. S-G, Table S7).

3.3. Performance Metrics

Because the classes are severely imbalanced, plain accuracy is misleading. We adopt macro-F1 (F1_m, the unweighted mean of per-class F1) as the primary metric, since it weights every crop equally and penalizes failure on the minority cereals where models differ most. We report three further metrics: Cohen's Kappa; weighted F1 (wF1, the support-weighted mean of per-class F1); and cross-entropy (Xent), the field-level scoring metric of the AI4FoodSecurity challenge (AI4EO, 2021) from which the dataset is drawn (lower is better, computed on hard label predictions). Formal definitions are in the Supplementary Material (Sec. S-B).

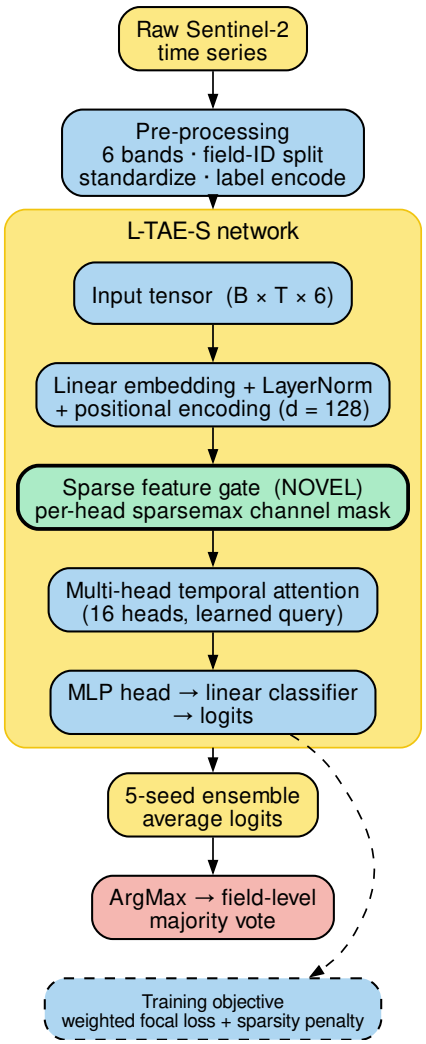


Figure 2: Architecture of the sparse-attention L-TAE-S. Before the multi-head temporal attention, a TabNet-style sparse feature gate (highlighted) applies a per-head `sparsemax` mask that selects a small subset of embedding channels, with a relaxation prior ($\gamma = 1.5$) discouraging heads from re-selecting the same channels. This block is the only architectural difference from the L-TAE; five seeds are ensembled and aggregated to field level.

3.4. Quantifying Uncertainty

We quantify two sources of variability. For sampling uncertainty in the finite holdout, we apply a nonparametric bootstrap over the 2,417 holdout fields: for each model we resample its per-field predictions with replacement $B = 10{,}000$ times and recompute macro-F1, reporting the 2.5th–97.5th percentile interval as a 95% confidence interval (CI). For between-model differences, we use a paired bootstrap: each replicate resamples the holdout fields once and scores both models on the same fields, giving 95% CIs for Δ macro-F1, $\Delta\kappa$, and Δ accuracy; an exact McNemar test on per-field correctness serves as a check. For optimization noise, the per-seed standard deviation of OOR macro-F1 across the five ensemble members is 0.015–0.037 for the headline temporal models, which is why we report five-seed ensembles throughout. Both sources are small relative to the ranking effects below: every field-bootstrap CI spans only $\approx\pm$0.02–0.03 macro-F1, so single-model intervals separate only for differences of $\approx$0.05 or more. Specific comparisons are therefore tested with the paired bootstrap, which is more sensitive because field difficulty cancels.

4. Results

We evaluate every model under two protocols: 1) in-region field-wise validation and 2) out-of-region transfer (OOR) to the adjacent, non-overlapping holdout tile. All reported metrics are field-level. Pixel- and patch-level models are aggregated to field labels, whereas field-level models are trained and evaluated on field-aggregated inputs

4.1. In-Region Validation Overstates Performance and Misranks Models

Table 1 reports both protocols for all models, ranked by OOR macro-F1, with the gap $\Delta = \mathrm{F1}_m^{\mathrm{OOR}} - \mathrm{F1}_m^{\text{in-region}}$. Two facts stand out. First, *every* model loses accuracy under transfer, but unevenly: from -0.04 (field-level L-TAE-S) to -0.24 (L-TAE at the pixel level). Second, this unevenness reorders the models. In-region the dense temporal pixel nets cluster at the top (L-TAE 0.78, Transformer 0.78, L-TAE-S 0.77, TempCNN 0.77, CNN-BiLSTM 0.72); a practitioner choosing by in-region F1 would pick from this cluster. On the holdout the cluster fractures: the Transformer drops to 0.58 and TempCNN to 0.57 ($\Delta = -0.20$), CNN-BiLSTM to 0.51 ($\Delta = -0.21$), and the in-region leader L-TAE ($\Delta = -0.24$) to 0.54, below every pixel-level tree learner.

The winners under transfer are a different group. The best single model is the TabNet pixel ensemble ($\mathrm{F1}_m = 0.60$, $\kappa = 0.54$), which also posts the lowest holdout cross-entropy (4.42) of any model, tied on macro-F1 by both sparse-attention L-TAE-S variants (0.60). The models with the smallest gaps are the sparse and linear learners: field-level L-TAE-S ($\Delta = -0.04$), logistic regression (-0.05), and the pixel-level tree learners. A small gap alone is not evidence of robustness, because it can also come from lower in-region accuracy: field-level L-TAE-S has the smallest gap largely because aggregation lowers its in-region macro-F1 from 0.77 to 0.64 while its holdout score stays at 0.60. We

therefore read gaps together with holdout accuracy. On that joint view, L-TAE-S, a deep temporal attention network whose only change is a learned gate restricting each head to a small channel subset, ties TabNet for the best holdout accuracy at both the pixel and the field level, which supports our thesis. The paired bootstrap confirms which differences are real: the best transferers beat the dense pixel nets by wide, significant margins (TabNet over CNN-BiLSTM, $\Delta = 0.09$, 95% CI $[0.07, 0.11]$; over L-TAE, $\Delta = 0.06$, $[0.04, 0.09]$), so the inversion is not holdout noise; at the very top the leaders (TabNet and both L-TAE-S variants) are statistically indistinguishable (all 0.60; TabNet vs. L-TAE-S, $p = 0.73$). We therefore do not claim a single best model; we claim that a sparse-feature-selection group occupies the top of the OOR ranking while the dense temporal nets prized by in-region validation fall significantly below it.

Table 1: In-region versus out-of-region transfer performance of all twenty-three models.

			In-reg		**Out-of-region (OOR)**				
Model	**Level**	**Features**	**F1$_m$**	**Δ**	**F1$_m$**	**95% CI**	**κ**	**wF1**	**Xent**
TabNet (raw)	pixel	raw	0.73	−0.13	**0.60**	.58–.63	**0.54**	0.70	**4.42**
L-TAE-S	field	raw	0.64	**−0.04**	0.60	.58–.62	0.52	0.69	4.77
L-TAE-S	pixel	raw	0.77	−0.17	0.60	.57–.62	0.53	**0.70**	4.63
L-TAE	field	raw	0.67	−0.07	0.59	.57–.62	0.47	0.67	5.51
Transformer	pixel	raw	**0.78**	−0.20	0.58	.55–.60	0.50	0.69	4.94
TempCNN	pixel	raw	0.77	−0.20	0.57	.55–.60	0.49	0.68	4.96
XGBoost	field	`xr_fresh`	0.70	−0.13	0.56	.54–.59	0.47	0.68	4.89
LightGBM	pixel	`xr_fresh`	0.62	−0.06	0.56	.54–.59	0.47	0.66	4.85
Logistic Regression	pixel	`xr_fresh`	0.61	−0.05	0.56	.54–.59	0.47	0.67	4.93
LightGBM	field	`xr_fresh`	0.68	−0.12	0.56	.53–.58	0.45	0.66	4.99
Voting	pixel	`xr_fresh`	0.69	−0.13	0.56	.53–.58	0.48	0.66	4.85
Random Forest	pixel	`xr_fresh`	0.65	−0.10	0.55	.52–.57	0.45	0.65	5.07
XGBoost	pixel	`xr_fresh`	0.64	−0.09	0.54	.52–.57	0.47	0.66	5.01
L-TAE	pixel	raw	**0.78**	−0.24	0.54	.51–.56	0.40	0.62	6.42
TempCNN	field	raw	0.67	−0.14	0.54	.51–.56	0.43	0.64	6.07
Stacking	field	`xr_fresh`	0.64	−0.11	0.53	.51–.56	0.48	0.67	5.32
CNN-BiLSTM	pixel	raw	0.72	−0.21	0.51	.49–.54	0.39	0.61	6.69
CNN-BiLSTM	field	raw	0.64	−0.14	0.51	.48–.53	0.33	0.54	7.86
3D CNN	patch	raw	0.70	−0.20	0.50	.48–.53	0.29	0.54	7.98
SMOTE Stacked	field	`xr_fresh`	0.69	−0.21	0.49	.46–.51	0.43	0.63	5.37
Multi-Channel 2D CNN	patch	raw	0.63	−0.18	0.45	.42–.47	0.30	0.54	7.62
TabNet (raw, field-avg)	field	raw	0.62	−0.19	0.43	.41–.46	0.31	0.56	7.35
TabNet (`xr_fresh`)	field	`xr_fresh`	0.51	−0.15	0.37	.34–.40	0.25	0.53	6.13

Ranked by OOR macro-F1. In-region uses field-wise validation; OOR uses the disjoint holdout tile (`34S_20E_259N`). Δ is the gap; the 95% CI is the percentile bootstrap on OOR macro-F1 from 10,000 resamples of the 2,417 holdout fields. Xent is the AI4FoodSecurity field-level scoring metric (lower is better). *Features* marks the automated `xr_fresh` statistics (174 features) versus raw reflectance. Bold marks the best value per column.

4.2. Pixel Versus Field Level: Aggregation as Variance Reduction

Several architectures were run at both the pixel and the field level (averaging per-pixel sequences within a field before classification). Table 2 shows that field-level aggregation narrows the gap of every *dense* temporal network (L-TAE $-0.24 \rightarrow -0.07$, TempCNN $-0.20 \rightarrow -0.14$, CNN-BiLSTM $-0.21 \rightarrow -0.14$),

Table 2: Pixel- versus field-level out-of-region transfer.

	Pixel		Field		
Architecture	**F1$_m$**	**Δ**	**F1$_m$**	**Δ**	**Field input**
L-TAE-S	**0.60**	−0.17	**0.60**	**−0.04**	raw-avg
L-TAE	0.54	−0.24	0.59	−0.07	raw-avg
XGBoost	0.54	−0.09	0.56	−0.13	`xr_fresh`
LightGBM	0.56	**−0.06**	0.56	−0.12	`xr_fresh`
TempCNN	0.57	−0.20	0.54	−0.14	raw-avg
CNN-BiLSTM	0.51	−0.21	0.51	−0.14	raw-avg
TabNet	**0.60**	−0.13	0.43	−0.19	raw-avg

Architectures run at both granularities, ranked by field-level OOR F1$_m$. The *Field input* column gives the field-level input (`xr_fresh` statistics for the trees, field-averaged raw sequence for the deep models). Field aggregation narrows the gap for the dense temporal nets but raises their OOR F1$_m$ only for L-TAE, is roughly neutral for the tree learners, and degrades TabNet. Bold marks the best value per column.

but it raises holdout accuracy only for L-TAE (F1$_m$ 0.54 → 0.59; paired $\Delta = 0.05$, 95% CI $[0.03, 0.08]$). For TempCNN (0.57 → 0.54) and CNN-BiLSTM (0.51 → 0.51) the gap closes mainly because in-region accuracy falls, so aggregation acts as a variance-reduction regularizer that trades in-region fit for stability rather than a reliable accuracy gain. For L-TAE-S the transfer gap shrinks from −0.17 to −0.04, but OOR F1$_m$ stays at 0.60, so here too aggregation stabilizes transfer rather than improving out of region performance. To demonstrate that more recent time series methods follow these same patterns we added two baselines: TimesNet and MiniRocket. For TimesNet aggregation lowers OOR F1$_m$ (0.57 → 0.52) and MiniRocket (0.50 → 0.48) (Supplementary Material, Sec. S-I). TabNet is the informative exception: aggregating its raw inputs to the field level *hurts* (F1$_m$ 0.60 → 0.43), because its robustness already derives from sparse feature selection and field-level training starves it of the per-pixel observation volume it exploits. The tree learners transfer well at either level, and their pixel→field change is essentially neutral under transfer (XGBoost 0.54 → 0.56). In short, aggregation stabilizes dense models but raises their transfer accuracy only for L-TAE; sparse, tree-like models do not need it.

4.3. Per-Class Performance and Persistent Confusions

Figure 3 breaks accuracy down crop by crop on the holdout for four representative models: the best transferer (TabNet), our sparse-attention L-TAE-S, its dense counterpart L-TAE, and CNN-BiLSTM (the weakest pixel-level temporal network on the holdout). The errors are highly uneven. TabNet and L-TAE-S classify the dominant lucerne/medics and the distinctive canola well (F1 0.88 and 0.81; 0.87 and 0.76), L-TAE less so (0.77 and 0.67), but every model struggles with the cereals and small-grain grazing: even TabNet reaches only 0.55 on barley, 0.48 on wheat, and 0.30 on small-grain grazing. Adding the sparsity gate (L-TAE → L-TAE-S) on the same backbone improves every class, most on the easier ones (lucerne/medics 0.77 → 0.87, canola 0.67 → 0.76)

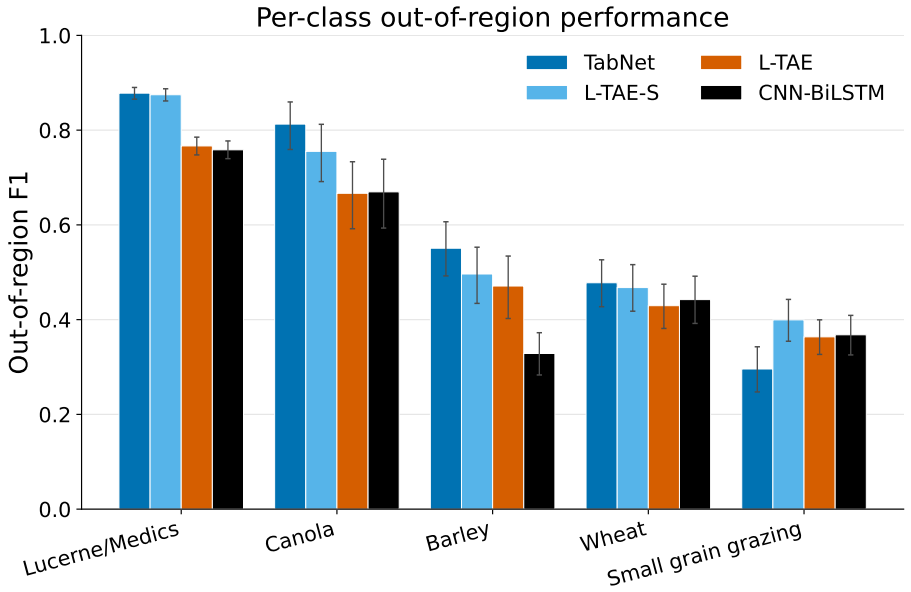


Figure 3: Per-class F1 on the holdout region for four representative models (TabNet, L-TAE-S, L-TAE, and CNN-BiLSTM). TabNet and L-TAE-S classify lucerne/medics and canola best; the cereals (barley, wheat) and small-grain grazing concentrate the errors. The dense L-TAE and CNN-BiLSTM also trail on the easier classes. Error bars are 95% bootstrap CIs over holdout fields.

and modestly on the cereals (barley 0.47 → 0.50, wheat 0.43 → 0.47, small-grain grazing 0.36 → 0.40). CNN-BiLSTM tracks L-TAE on most classes but is weakest on barley (0.33), so the dense networks lose accuracy on easy and hard classes alike, not only on minority classes. The wheat–barley confusion is consistent with their spectral overlap and appears across every model. It appears that errors are also spatially clustered and do not degrade with distance. Close to the training tiles, these four models and field-level XGBoost each classify about 70% of fields correctly; however, in the south-central zone (about 8 to 16 km away) L-TAE and CNN-BiLSTM fall to 36% and 30% while TabNet and L-TAE-S stay near 70% (Supplementary Material, Sec. S-E, Fig. S7).

4.4. Data Efficiency Under Out-of-Region Transfer

The most transferable models are also the most data-efficient. We retrain a representative set on random 75%, 50%, and 25% subsets of the training fields and re-evaluate on the holdout; because a single subsample can be unrepresentative, we repeat each fraction over five independent draws and plot the mean (Fig. 4). Averaged this way, the deep temporal models are essentially flat from full data down to a quarter of the fields (L-TAE and L-TAE-S vary by under 0.02 macro-F1 across fractions), while the tree, linear, and TabNet baselines lose only ∼0.03–0.05, modest given a four-fold cut in training data. The per-fraction

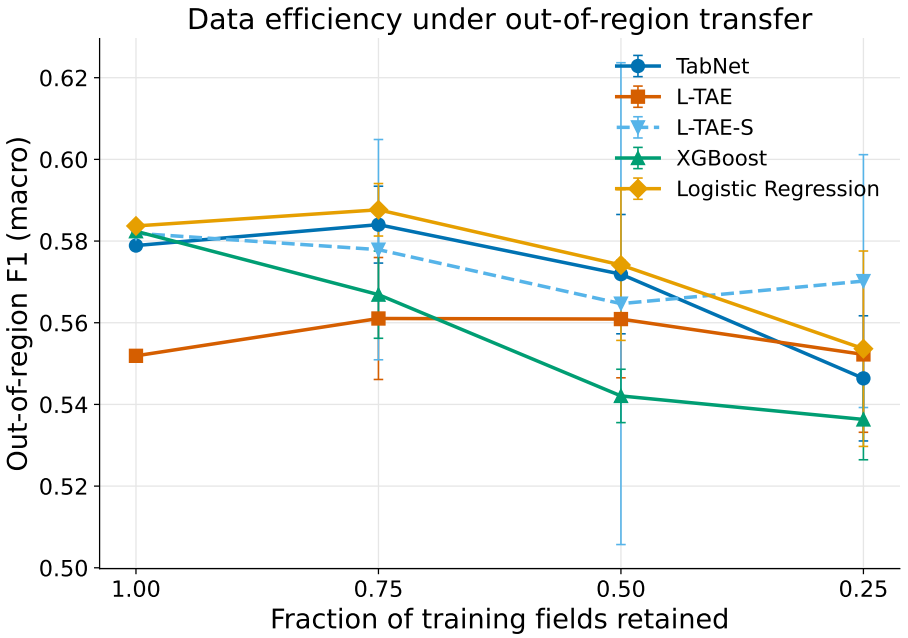


Figure 4: Data efficiency under out-of-region transfer: macro-F1 versus fraction of training fields retained, single training seed, each point the mean over five independent subsample draws (error bars ±1 s.d.). The deep temporal models (L-TAE, L-TAE-S) are nearly flat to a quarter of the fields; the tree, linear, and TabNet baselines decline modestly. Per-fraction variation is within draw-to-draw sampling noise, largest for L-TAE-S (±0.06 at 50%), so single-split "dips" are not real. Absolute values run ∼0.02–0.05 below the five-seed levels of Table 1 because a single training seed is used.

wiggles seen in any single split are within draw-to-draw sampling variance: L-TAE-S, whose sparse channel gate makes it the most sensitive to *which* fields are sampled, swings ±0.06 macro-F1 at the 50% fraction (versus ≤ 0.02 for the robust learners), so the apparent 50% "dip" of any one draw is sampling noise rather than a data-efficiency effect. Holdout accuracy is therefore limited far more by out-of-region transfer than by training-set size, reinforcing that what governs holdout accuracy is the model's inductive bias rather than the volume of training data. Because each subsample is drawn across the full training extent, this experiment varies the labeled training area (from ≈620 to ≈155 km^2 of fields) rather than the spatial extent of the training region.

4.5. Raw Reflectance versus `xr_fresh`

The best-transferring classical learners in Table 1 are exactly those fed `xr_fresh` statistics, so representation and model family are confounded there. To separate them, we trained a single XGBoost learner once on raw monthly reflectance and once on the per-pixel `xr_fresh` statistics, holding the FID-wise splits, preprocessing, and field-level mean pooling fixed (Table 3). The two rep-

Table 3: Raw reflectance versus `xr_fresh` for a single fixed XGBoost learner.

Representation	**Split**	**F1$_m$**	**wF1**	κ
Raw reflectance	In-region	0.73	0.80	0.71
`xr_fresh`	In-region	0.72	0.76	0.68
Raw reflectance	Out-of-region	0.59	0.69	0.50
`xr_fresh`	Out-of-region	0.56	0.69	0.50

Identical FID-wise splits, preprocessing, and field-level mean pooling; only the input representation differs. Metrics are field-level macro-F1 (F1$_m$), weighted F1 (wF1), and Cohen's κ.

resentations are essentially indistinguishable under transfer (F1$_m$ = 0.59 raw vs. 0.56 `xr_fresh`, a difference within the holdout bootstrap interval of Table 1; equal weighted-F1 and κ), and raw reflectance is, if anything, marginally ahead in-region as well. Field-level TabNet, run with the same configuration and seeds on both inputs, points the same way: field-averaged raw sequences transfer better than `xr_fresh` (OOR F1$_m$ = 0.43 vs. 0.37, non-overlapping bootstrap intervals; Table 1). For these two learners, `xr_fresh` yields no transfer-accuracy gain over raw reflectance. Its value is operational: a compact, fast-to-fit, agronomically interpretable vector that trains in seconds (Supplementary Material, Sec. S-C). The comparison does not extend to the deep temporal models, because `xr_fresh` reduces each time series to summary statistics with no time axis for L-TAE, TempCNN, or CNN-BiLSTM to attend or convolve over. This is, to our knowledge, the first controlled comparison of `xr_fresh` against raw reflectance under out-of-region transfer.

5. Discussion

The conclusion about which model is "best" depends almost entirely on whether evaluation is in-region or under out-of-region transfer, and a model's transfer behavior is predictable from its inductive bias.

5.1. Why Tree-Like Inductive Bias Transfers

The clearest pattern is that transfer is strongly associated with inductive bias, and not simply with model family or attention capacity. Figure 5 groups every model by inductive bias and plots its OOR gap: linear models lose the least (mean $\Delta = -0.05$), tree-based models little (mean -0.11), the sparse-attention L-TAE-S sits essentially with the trees (mean -0.11 across its variants), while dense temporal/patch nets lose the most (mean -0.20) and SMOTE-augmented stacking is similarly fragile (-0.21). RF and gradient-boosted trees have been the remote-sensing workhorse for two decades not because they top in-region accuracy (dense nets often do) but because they survive when a model is applied to an unseen region (the operational case) (Belgiu and Drăguţ, 2016; Maxwell et al., 2018). A tree decides one feature at a time, choosing the few features that matter for each split and ignoring the rest; this sparsity may reduce reliance on region-specific quirks of phenology, soil, and atmosphere that pull richer models

off course. Independent corroboration spans settings: a 3D-CNN collapses under cross-sensor hyperspectral transfer while a sparse SVM holds (Pankajakshan et al., 2025); RF matches or exceeds a dense CNN-LSTM on label-scarce Ghana fields (Rustowicz et al., 2019); a maize model drops 16 points moving across East Africa (Jin et al., 2019); and bagged/boosted trees out-transfer an elaborate instance-reweighting scheme from the U.S. to northwest China (Mai et al., 2025). TabNet, the best transferer here, is a neural network built to behave like a tree: a small learned gate picks a handful of features at each step (Arik and Pfister, 2021), so it clusters with the trees and inherits their transferability. By contrast, CNN-BiLSTM, TempCNN, L-TAE, and the Transformer consume the whole temporal sequence at once with no feature selection. At the pixel level this leaves them fitting the training region tightly: they post the largest in-region-to-holdout gaps in the benchmark, with L-TAE dropping furthest ($\Delta = -0.24$). This is consistent with evidence that the dates such models attend to are themselves contingent on the training region (Obadic et al., 2022). Dense attention is not intrinsically poor under out-of-region transfer: the pixel-level L-TAE loses 0.24 macro-F1, but the field-level L-TAE reduces this gap to 0.07 and raises holdout macro-F1 to 0.59 (Sec. 4.2), so it is the pixel-level dense models specifically that transfer worst.

The Transformer and L-TAE-S together separate attention capacity from feature selection on the same pixel-level task. The Transformer deploys far richer attention than L-TAE's single query and does transfer better ($F1_m$ 0.58 vs. 0.54; paired $\Delta = 0.04$, 95% CI $[0.02, 0.06]$), so capacity is not irrelevant. Adding sparse feature selection to the smaller L-TAE helps more: L-TAE-S reaches 0.60 ($\Delta = 0.06$, $[0.04, 0.08]$; $\Delta\kappa = 0.13$) and matches or exceeds the Transformer (Δ macro-F1 0.02, $p = 0.06$; $\Delta\kappa$ and Δ accuracy 0.02, both significant). At the field level the gate's macro-F1 gain over L-TAE is not significant (0.60 vs. 0.59), but κ and accuracy improve ($\Delta = 0.04$ and 0.05, both significant). L-TAE-S and TabNet, the two models that explicitly select features before computing with them, are the best transferers. Within the architectures tested, sparse feature selection is therefore associated with larger and more consistent transfer gains than added attention capacity; this is consistent with, but does not prove, feature selection as the property that matters most. Two recent time-series classifiers point in a similar direction (Supplementary Material, Sec. S-I). TimesNet (Wu et al., 2023), a high-capacity network, its out-of-region transfer is similar to Transformer and TempCNN ($F1_m = 0.57$). MiniRocket (Dempster et al., 2021), which passes 10,000 unselected random-convolution features to a linear classifier, performs worse than logistic regression on the raw inputs (0.50 vs. 0.56). Adding the L-TAE-S style gate to MiniRocket's features raises its holdout macro-F1 by 0.06 over an identical network with a dense gate, the same gain the gate gives L-TAE. The L-TAE-S experiment also shows that this is *not* a tree-versus-deep tradeoff: the same selection step used by tree splits can be grafted onto a dense architecture at the cost of a single block, and L-TAE-S ties TabNet for the best holdout accuracy. This result does not hinge on the chosen hyperparameters: across 22 of 23 field-level settings of γ and the number of heads, L-TAE-S scores between 0.58 and 0.62, and at

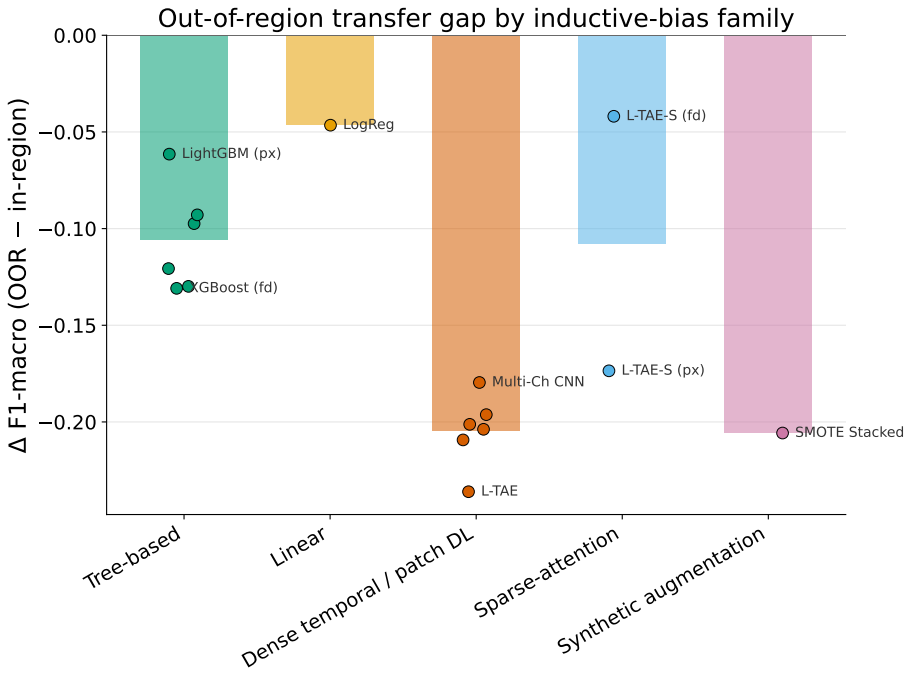


Figure 5: Out-of-region transfer gap (Δ macro-F1) grouped by inductive-bias family. Sparse/tree-like and linear models transfer (small $|\Delta|$); dense temporal/patch nets and synthetic augmentation overfit the training region. L-TAE-S, which adds sparse feature selection to a dense attention backbone, tracks the tree-based models. Bars are family means; points are individual models.

both pixel-level settings tested it still improves significantly on L-TAE, although fewer heads transfer slightly better (Supplementary Material, Sec. S-G). That parity is not accuracy alone: at the pixel level L-TAE-S reaches TabNet's best-in-class holdout accuracy in roughly 40% of the per-seed training time (24 vs. 61 min), and its field-level variant is both faster and more accurate than any field-aggregated TabNet (Supplementary Material, Sec. S-C). The next generation of operational crop classifiers need not choose between the expressivity of deep learning and the transfer-robustness of trees.

5.2. Two Robustness Levers

Two caveats sharpen rather than weaken the story, and together they point to how we can choose the best models. First, it is *pixel-level* TabNet that wins; the field-aggregated TabNet variants are among the worst transferers (Table 1), because aggregation removes the per-pixel observation volume the sparse model exploits. Second, field-level aggregation helped only one dense model: L-TAE's transfer gap shrinks from -0.24 to -0.07 and its holdout $F1_m$ rises to 0.59, whereas for TempCNN, CNN-BiLSTM, TimesNet, and MiniRocket it narrows

the gap without raising, or while lowering, holdout accuracy (Table 2; Supplementary Material, Sec. S-I). These point to two directions to improve transfer accuracy albeit with unequal performance. The first approach is built on feature selection. The model itself picks a few features at a time (trees, TabNet, and now L-TAE-S). The second is variance reduction imposed from outside via field-level aggregation, which averages away region-specific pixel noise before the classifier sees it. It is available even when the architecture is fixed, but its effect on accuracy depends on the model. A model that gains with the first lever (sparsity) may degrade with the second lever (aggregation)because of the loss observations; this is exactly why TabNet prefers pixels. However, these two enhancements can be combined. L-TAE-S adds a sparse gate to a temporal-attention backbone and then tolerates field aggregation: its holdout macro-F1 stays at 0.60 while per-seed training time falls from 24 min to 18 s. Practitioners can read this as a flowchart: if free to choose or modify the model, build feature selection in (lever 1); if constrained to an off-the-shelf dense architecture, test field-level aggregation (lever 2), which stabilizes transfer but raised accuracy for only one of the five dense models we ran at both levels. Two further controlled comparisons confirm the ranking is not an artifact of ensembling or imbalance handling; both shift absolute scores but never reorder the robust or fragile groups, see Supplementary Material (Sec. S-D). Seed-ensembling gives only a small, monotone lift that leaves the ordering intact. A single-seed L-TAE-S ($F1_m = 0.56$) already exceeds the five-seed CNN-BiLSTM and TabNet field ensembles (Table S4). And the imbalance-handling objective moves absolute scores but not the ranking: the robust L-TAE-S is almost invariant across five loss and sampler settings ($F1_m = 0.59$–0.60), whereas the fragile CNN-BiLSTM swings widely (0.41–0.56) and even at its best stays below L-TAE-S's worst (Table S5).

5.3. Automated Features, Patch Models, and Persistent Difficulty

The classical pipelines paired with `xr_fresh` were not only robust but cheap, transferring as well as the best deep models (OOR $F1_m$ 0.55–0.56) while training in minutes rather than hours (Supplementary Material, Sec. S-C); the controlled comparisons for XGBoost and TabNet (Sec. 4.5) show that this value is operational rather than an accuracy edge over raw reflectance. Patch-based convolutional models did not exceed pixel-level models and transferred poorly (3D CNN $F1_m = 0.50$; multi-channel 2D CNN 0.45). We tested one configuration of each, so this should not be read as a property of patch CNNs in general. Several design factors may also contribute to it. At 10 m resolution a 100 m patch holds only $\sim 10 \times 10$ pixels, and upsampling to 128×128 adds interpolation rather than detail. After upsampling, the convolutional receptive fields span only one to two native pixels before global pooling or flattening, so little spatial structure is learned. The two models also fuse time differently: the 2D CNN stacks all band-month images as input channels, discarding temporal order, while the 3D CNN convolves across months. Lack of sparse feature selection may also play a role. Our experiments do not separate these factors; patch-size and architecture sweeps would be needed to do so. Finally, small-grain grazing,

barley, and wheat are the hardest classes for every model (Fig. 3) and account for most accuracy lost under transfer; that even robust models struggle here, and that the dense networks trail on the easier classes as well, indicates the limitation is partly intrinsic spectral similarity and partly the transfer problem itself, not class imbalance alone. Much of this confusion is among the cereals themselves rather than with the other crops. Merging barley, wheat, and small-grain grazing into a single cereal class, without retraining, raises out-of-region accuracy from 0.73 to 0.84 for TabNet and from 0.71 to 0.83 for L-TAE-S, with the merged class reaching an F1 of about 0.77. Under the merged scheme the sparse models remain at the top of the ranking and the dense pixel networks at the bottom (Supplementary Material, Sec. S-F, Table S6). The cereals' seasonal profiles nearly coincide, and the later holdout season shifts them by more than they differ from one another, which adds to their confusion under transfer (Supplementary Material, Sec. S-H).

5.4. Scope and Practical Relevance

The absolute accuracies reported here (out-of-region macro-F1 of at most 0.60) are too low for an operational crop map, and producing one is not our aim. The study is a controlled experiment on how model families behave under the low-information conditions typical of much of Africa: no labels in the target area, a modest labeled training set from a single season, and optical imagery only. Under these constraints the useful result is relative: which models lose the least when moved to unlabeled land, and how to choose among them when no target labels exist to check that choice. Three practical points follow. First, a model's accuracy within the training region is a poor guide to how it will perform elsewhere, so when no labels exist in the target area, the safer choice is a model with built-in sparse feature selection, such as TabNet or L-TAE-S, or a dense model whose predictions are aggregated to the field level. Second, the crop-specific results are more useful than the overall score. Lucerne/medics, the dominant crop, and canola transfer well: TabNet and L-TAE-S reach out-of-region F1 of 0.87 to 0.88 for lucerne/medics and 0.76 to 0.81 for canola, so these crops can be mapped with reasonable confidence in unlabeled neighboring areas. The cereals do not: barley (0.50 to 0.55), wheat (0.47 to 0.48), and small-grain grazing (0.30 to 0.40) remain confused with one another for every model, consistent with their overlapping spectral and phenological profiles. In this setting cereal maps are better reported as a combined cereal class, or improved with additional data such as SAR, than separated with optical time series alone. The sparse gate helps across all five crops (L-TAE to L-TAE-S), but it does not close the cereal gap. Third, the robust models are cheap to train: the classical pipelines need minutes and field-level L-TAE-S seconds per seed, which suits settings with limited computing resources. Adding SAR, multiple seasons, or a small number of target-region labels is the natural route from this benchmark toward operational accuracy.

5.5. Limitations

The study covers one growing season (2017) and one agroecological zone, with a holdout tile *adjacent* to the training tiles. Holdout fields lie 0 to 24 km from the nearest training field (median 11 km). The models are trained on $\approx$620 km^2 of labeled fields within a $\approx 25 \times 45$ km region and are used to predict in a $\approx 24 \times 24$ km tile. This is a relatively local shift, so the gaps we report most likely *understate* the degradation under cross-regional or cross-year transfer and are best read as a conservative lower bound. Cross-year and cross-ecozone transfer could not be tested, because the benchmark labels only these three tiles in a single season. Importantly, in South Africa, like much of Africa, training data is sparse if not non-existant. Multi-year, multi-region holdouts are needed to confirm that the inductive-bias ordering holds under larger shifts. Our conclusions therefore apply to out-of-region transfer between neighboring tiles at these distances and scales. Training-set size is the one scale we vary: the most transferable models keep near-full holdout accuracy on a quarter of the training fields (Sec. 4.4). The labels derive from a public dataset rather than our own campaign, and we use Sentinel-2 optical data only; a deliberate scoping choice that nonetheless limits the achievable ceiling for optically similar cereals, making SAR fusion a clear next step.

6. Conclusion

In-region validation, the protocol the crop-classification literature reports almost universally, misranks twenty-three Sentinel-2 classifiers when they are applied to an adjacent, non-overlapping holdout tile. Dense temporal networks that lead under conventional evaluation fall sharply under transfer, the in-region leader (L-TAE) dropping below every pixel-level tree learner, while parsimonious models such as tree ensembles, logistic regression, and a TabNet ensemble retain rank and accuracy. The common thread is the same property that made tree ensembles the remote-sensing default for two decades: a sparse, axis-aligned feature-selection bias that commits to a few robust signals and ignores the rest. TabNet inherits it through sparsemax masks. A Transformer with far greater attention capacity transfers better than the lightweight L-TAE, but adding a sparse gate to L-TAE helps more, pointing to sparsity rather than capacity as the stronger lever. Grafting that lever onto an attention model (L-TAE-S) ties TabNet for the best transfer at a fraction of its training cost and, at the field level, keeps that accuracy while training in seconds per seed. Field-level aggregation narrows the transfer gap of dense temporal networks, but it raised holdout accuracy only for L-TAE and lowered it for TabNet and several other dense models. We therefore recommend the following: evaluate on an out-of-region holdout before declaring one architecture better than another; prefer models with a built-in feature-selection bias when mapping nearby unlabeled areas; and, if dense temporal models are required, test field-level aggregation rather than assume it helps. The conditions that make the Western Cape difficult are the conditions under which agricultural remote sensing must operate

across much of the Global South, and evaluating models the way they are actually deployed, applied to regions where no labels were collected, is what will make remote sensing deliver where it matters most.

CRediT authorship contribution statement

Michael L. Mann: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Sairam Venkatachalam:** Software, Formal analysis, Writing – original draft, Investigation. **Disha Kacha:** Software, Formal analysis, Investigation, Writing – original draft. **Devarsh Sheth:** Software, Formal analysis, Investigation, Writing – original draft. **Ryan Engstrom:** Supervision, Methodology, Writing – review & editing. **Amir Jafari:** Methodology, Methodology, Supervision, Writing – review & editing.

Ethics statement

This article does not contain any studies with human participants or animals performed by any of the authors. The study is based solely on publicly available satellite imagery (Sentinel-2) and field-boundary crop labels, so no ethical approval was required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The processed dataset supporting this study is openly available on IEEE DataPort at `https://dx.doi.org/10.21227/n69k-r458` (DOI: 10.21227/n69k-r458). The source code is available at `https://github.com/mmann1123/South_Africa_Crop_Comp` (Mann, 2025a). The study builds on the ground-reference crop labels of the Western Cape Department of Agriculture and Radiant Earth Foundation, "Crop Type Classification Dataset for Western Cape, South Africa", v1.0, Radiant MLHub (DOI: 10.34911/rdnt.j0co8q), used under the CC-BY-4.0 license; Sentinel-2 imagery was extracted via Google Earth Engine.

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